

Toward One-Step Diffusion Language Models via Average-Velocity Distillation on the Probability Simplex

Anonymous submission

Abstract

Diffusion language models (DLMs) promise parallel, bidirectional text generation, yet in practice they remain slow: matching autoregressive quality requires tens to hundreds of denoising steps. In the image domain, one-step generation is now routine, but the leading construction—a learnable *flow map*—has been argued to be unavailable to discrete masked diffusion, the paradigm behind the largest open DLMs. We revisit this claim. We introduce SiFLOW, a framework that distills a *pretrained* masked DLM into a one- or few-step generator by learning an *average unmasking velocity* on the probability simplex of the teacher’s predictive distribution. SiFLOW adapts the average-velocity identity of MeanFlow (Geng et al. 2025) to the discrete setting, and we show that the resulting secant target is the constant-velocity field of a straight-line path on the simplex—the discrete analog of the straightened conditional paths used in Flow Matching and OT-CFM (Lipman et al. 2023; Tong et al. 2024). Unlike concurrent work that requires training from scratch (Yoo et al. 2026), SiFLOW pairs this construction with a temperature-annealed distillation objective and an entropy-injected prior to distill any existing pretrained masked DLM—including MDLM, Dream-7B, and DiffusionGemma, each on its native tokenizer—without modifying their architectures. We give a self-contained construction, a first-order error analysis, and a reproducible experimental protocol designed to run on a single GPU, covering three open DLM families and isolating the contribution of each component on standard language-modeling benchmarks.

1 Introduction

Autoregressive (AR) language models generate one token at a time, so producing L tokens costs L sequential forward passes. This left-to-right dependency caps latency and underuses parallel hardware, since decoding is bound by memory bandwidth rather than compute. Diffusion language models (DLMs) offer a structurally different route: starting from a fully masked or noised sequence, they update all positions at once under bidirectional attention (Austin et al. 2021; Lou, Meng, and Ermon 2024; Sahoo et al. 2024). At scale, masked DLMs such as LLaDA (Nie et al. 2025), Dream (Ye et al. 2025), and DiffusionGemma (Google DeepMind 2026) now rival similarly sized AR models, with the last achieving over 1,000 tokens per second on a single H100.

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The promise of parallel generation, however, is undercut by iterative refinement. A masked DLM commits reliable tokens only a few at a time, so high-quality samples still require many *number of function evaluations* (NFEs)—often comparable to the sequence length (Nie et al. 2025; Deschenaux and Gulcehre 2025).¹ The field has turned to acceleration: consistency-style distillation and few-step samplers reduce the step count, but for discrete DLMs they typically bottom out at a handful of steps rather than one (Deschenaux and Gulcehre 2025; Zhu et al. 2025a; Yoo et al. 2026; Zhang et al. 2026).

This stands in sharp contrast to image generation, where one-step models match their multi-step teachers. MeanFlow attains competitive single-step quality by learning an *average velocity* field rather than an instantaneous one (Geng et al. 2025), and distribution-matching distillation collapses a diffusion teacher into a one-step student (Yin et al. 2024; Song et al. 2023). The natural question is whether the same is possible for language. Three obstacles are documented in the literature. (i) *Entropy collapse*: a single pass must place near-deterministic mass on each token, a target that masked-diffusion training never directly supervises (Zhu et al. 2025b). (ii) *Token independence*: parallel sampling draws each position from its own marginal, so without iterative correction, individually plausible tokens combine into incoherent sequences. (iii) *No flow map*: unlike continuous diffusion, discrete masked diffusion does not admit a unique noise-to-data map that could serve as a direct one-step regression target.

The third obstacle has motivated a continuous detour. Flow Map Language Models (FMLM) lift tokens to one-hot embeddings and run a continuous flow, which *does* admit a learnable flow map for few-step inference; the authors explicitly position this map as a structure “unavailable to discrete methods” (Lee et al. 2026). This is effective, but it forfeits compatibility with the pretrained masked DLMs that dominate at scale. Concurrent work IMDM (Yoo et al. 2026) avoids the discrete barrier by replacing the single mask token with an infinite-state stochastic mask, but likewise requires training from scratch and targets few-step rather than

¹NFE counts student forward passes, including the lightweight velocity-head pass; SiFLOW’s one-step generator therefore costs 1 NFE.

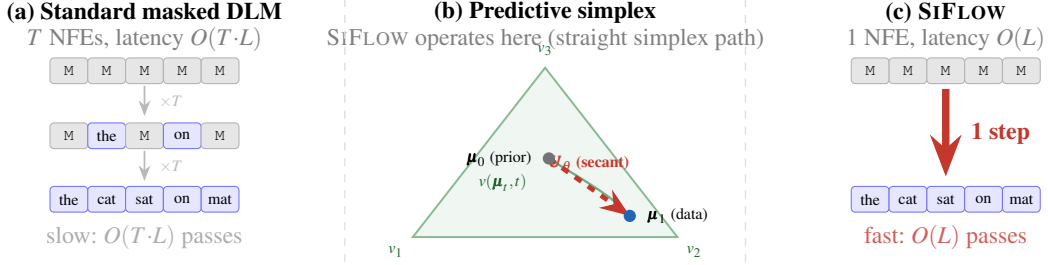


Figure 1: **Key idea of SiFLOW.** (a) Standard masked DLMs unmask tokens iteratively (T NFES). (b) A pretrained masked DLM emits, at every noise level, a predictive distribution μ_t over the vocabulary—a point on the probability simplex. This trajectory is smooth even though tokens are discrete, so an *average-velocity* shortcut $\mathbf{U}_\theta$ (red dashed) can approximate the full instantaneous curve v_t (green) in a single step. The secant target is the constant-velocity field of the straight simplex path (Section 3.2). (c) SiFLOW maps a fully masked prior to a complete sequence in one pass.

one-step generation.

We argue the discrete barrier is an artifact of where one looks for the flow map. Although the token state is discrete, a pretrained masked DLM emits, at every position and every noise level, a *predictive distribution over the vocabulary*—a point on the probability simplex. The trajectory of these predictive simplex points, as the noise level decreases, is a perfectly well-defined object on which an average-velocity construction applies. Crucially, the secant approximation of this average velocity is the *constant-velocity field of a straight path between simplex points*—the discrete analog of the straightened conditional paths that make few-step Flow Matching and OT-CFM efficient (Lipman et al. 2023; Tong et al. 2024). Operating on the simplex rather than in token space recovers a flow-map analog for discrete masked diffusion while keeping the discrete teacher intact.

We introduce SiFLOW, a distillation framework that turns any pretrained masked DLM into a one- or few-step generator. SiFLOW defines an *average unmasking velocity* on the teacher’s predictive simplex, derives the corresponding consistency identity as a discrete analog of MeanFlow, and distills it into a student that adds only a lightweight velocity head to the teacher backbone. To make single-pass generation trainable, SiFLOW adds a temperature-annealed simplex distillation loss and an entropy-injected prior adapted from Di[M]O; a cheap self-conditioned refinement recovers token dependencies in a few steps when desired.

Plain MeanFlow assumes a continuous Euclidean state with a unique probability-flow ODE; discrete masked diffusion has no token-space flow, and naive distillation matches only the clean marginal, discarding the trajectory. SiFLOW’s contribution is identifying that the trajectory exists on the predictive simplex and that its average velocity—a secant chord of one nested-mask trajectory—is the regressable one-step target. This paper makes four contributions. First, we reframe one-step discrete diffusion as average-velocity learning on the predictive simplex (Figure 1), give a flow-map analog where prior work assumed none, and connect the construction to the straightened conditional paths of Flow Matching and OT-CFM (Section 3). Second, we instantiate it as SiFLOW, a distillation procedure compatible with off-the-shelf masked DLMs without retraining from

scratch (Figure 2; Section 3). Third, we provide a first-order error analysis relating one-step quality to the average-velocity approximation (Section 4). Fourth, we specify a reproducible, single-GPU experimental protocol that applies the same head-only recipe across three masked-DLM backbones (MDLM-170M, Dream-7B, DiffusionGemma-26B-A4B), each on its native tokenizer (Section 5).

2 Background and Related Work

Masked diffusion language models. Let $\mathcal{V}$ be a vocabulary of size K and $[M]$ an absorbing mask token. A masked DLM defines a forward process that, at noise level $\tau \in [0, 1]$, independently replaces each token by $[M]$ with probability $\sigma(\tau)$, where σ is increasing with $\sigma(0) = 0$ (clean) and $\sigma(1) = 1$ (fully masked). A bidirectional Transformer p_ϕ is trained to recover clean tokens from a masked state; with the SUBS parameterization and a Rao–Blackwellized objective, the loss reduces to a weighted sum of masked cross-entropies (Sahoo et al. 2024). SEDD reaches the same family through score-entropy ratios (Lou, Meng, and Ermon 2024), LLaDA scales the recipe to 8B parameters (Nie et al. 2025), Dream-7B initializes from an AR checkpoint for improved instruction following (Ye et al. 2025), block diffusion interpolates between AR and diffusion decoding (Arriola et al. 2025), and DiffusionGemma ships a 26B MoE as an open-weights production model (Google DeepMind 2026). At inference, the model iteratively predicts all positions and re-masks the least confident ones, so quality improves with the number of steps but so does latency.

One-step generative modeling. Consistency models enforce that points along a probability-flow trajectory map to a common origin, enabling single-step sampling (Song et al. 2023). Distribution-matching distillation trains a one-step generator whose output distribution matches a diffusion teacher (Yin et al. 2024). MeanFlow (Geng et al. 2025) is our central anchor: it characterizes a flow by its *average* velocity over an interval and derives an identity that supplies a regression target requiring no curriculum or distillation. OT-Conditional Flow Matching (Tong et al. 2024) shows that using minibatch-OT couplings to construct straight-line training paths dramatically reduces NFES at inference; our secant

target is the simplex analog of this coupling.

Acceleration and distillation for DLMs. Self-Distillation Through Time (SDTT) compresses a pretrained masked DLM from 1024 steps to fewer than 64 using reverse-KL targets (Deschenaux and Gulcehre 2025); divergence-instruct variants push discrete generation further (Zhu et al. 2025a). T3D (Zhang et al. 2026) combines trajectory self-distillation with a discriminative objective to reach 4–8 steps while retaining teacher quality; it operates on pretrained DLMs but does not reach one step. IMDM (Yoo et al. 2026) identifies a fundamental factorization error bound in standard MDMs and introduces a stochastic infinite-state mask to mitigate it, achieving state-of-the-art few-step results; crucially, it requires training from scratch and is evaluated at 4–8 steps, not one. DLM-One reaches a single step, but only for *continuous* DLMs, distilling the score of an embedding-space teacher (Chen, Zhang, and Zhou 2025). Di[M]O is the first to distill a *masked* diffusion model to one step, for images, and identifies the entropy-of-prior problem we inherit (Zhu et al. 2025b). FMLM achieves one-step *language* generation through a continuous flow on one-hot embeddings, arguing a usable flow map is unavailable to discrete methods (Lee et al. 2026). A complementary line of work accelerates discrete decoding without average-velocity distillation—training-free or plug-in samplers (adaptive unmasking with backtracking, dependency-guided parallel decoding, verifier-guided acceptance) and data–decoder co-design approaches (slot-level plan-and-infill, parallel-structured supervision); these are orthogonal to SiFLOW and could in principle be layered on top of a distilled student. SiFLOW differs on all three axes simultaneously from every prior method: it is one-step, operates in the discrete masked-diffusion paradigm, and distills from pretrained DLMs without architectural modification. Table 1 summarizes the landscape.

3 Method: SiFLOW

We first state where the flow map lives (Section 3.1), then define the average unmasking velocity and its OT connection (Section 3.2). Sections 3.3–3.5 address the two failure modes of single-pass generation, and Section 3.6 assembles the training procedure.

3.1 From Discrete Tokens to the Predictive Simplex

The state of a masked DLM is a discrete sequence, but its *output* is not. At position i and noise level τ , the teacher returns a categorical distribution

$$\mu_\tau^i = p_\phi(x_0^i | \mathbf{x}_\tau) \in \Delta^{K-1}, \quad (1)$$

a point on the $(K-1)$ -simplex. As τ decreases from 1 (fully masked) to 0 (clean), and as the teacher’s reverse process reveals context, this point moves: from a context-only prediction near the prior toward a near-vertex one-hot at the realized token. The induced curve $\tau \mapsto \mu_\tau^i$ is a well-defined trajectory on a smooth manifold, even though the underlying tokens are discrete. SiFLOW performs all of its geometry on

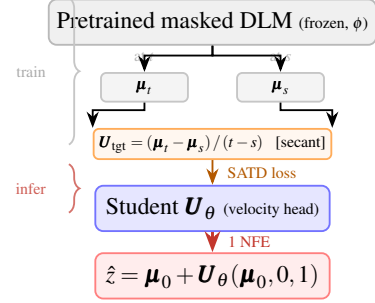


Figure 2: **SiFLOW training and inference.** The frozen pretrained DLM (any masked DLM family) provides predictive simplex points μ_t, μ_s at sampled noise levels $t > s$; their secant $\mathbf{U}_{\text{tgt}}$ —the constant-velocity field of the straight simplex path—regresses the student velocity head $\mathbf{U}_\theta$ via the SATD loss. At inference one forward pass maps the masked prior μ_0 to the predicted clean simplex.

the product of per-position simplices, rather than in token space. This is the key move: it sidesteps the non-existence of a token-space flow map by working with the object the model already produces, and it applies without modification to any masked DLM family—MDLM, Dream-7B, or DiffusionGemma.

To keep notation aligned with MeanFlow, we reverse the time axis for generation: write $s \in [0, 1]$ with $s = 0$ the fully masked prior and $s = 1$ the clean data, so generation runs $0 \rightarrow 1$. We suppress the position index i when clear from context and treat μ as the stacked simplex state over all positions.

Path consistency via nested masks. The two simplex states μ_s and μ_t that define a secant are not two unrelated conditional predictions at different corruption rates. We compute both from a *single* clean sequence $\mathbf{x}_1$ under a shared reveal order: the masked state $\mathbf{x}_t$ at the later (less-masked) level is obtained by revealing a superset of the positions revealed in $\mathbf{x}_s$, so $\text{mask}(t) \subseteq \text{mask}(s)$ (nested masks). Consequently μ_s and μ_t lie on one and the same reverse trajectory of the teacher, and their displacement quotient is a chord of that single trajectory rather than a difference of two independently sampled paths. This makes the induced curve $\tau \mapsto \mu_\tau$ a coherent flow line and is what justifies treating the secant as an average velocity (Section 3.2).

3.2 The Average Unmasking Velocity and OT Connection

Let $\mathbf{v}(\mu, s)$ denote the instantaneous velocity of the teacher-induced probability flow on the simplex. Following MeanFlow, we define the central object as a time-average.

Definition 1 (Average unmasking velocity) For $0 \leq s < t \leq 1$,

$$\mathbf{U}(\mu_s, s, t) := \frac{1}{t-s} \int_s^t \mathbf{v}(\mu_\tau, \tau) d\tau. \quad (2)$$

Because $\mathbf{U}$ is defined purely as an average of $\mathbf{v}$, differentiating the relation $(t-s)\mathbf{U} = \int_s^t \mathbf{v} d\tau$ in t (with s fixed)

Table 1: Where SiFLOW sits. “Pretrained DLM” = reuses an off-the-shelf model; “Discrete” = masked-token diffusion; “Domain” = image or text.

Method	One-step	Discrete	Pretrained DLM	Domain
Consistency (Song et al. 2023)	✓	—	—	image
MeanFlow (Geng et al. 2025)	✓	—	—	image
Di[M]O (Zhu et al. 2025b)	✓	✓	—	image
DLM-One (Chen, Zhang, and Zhou 2025)	✓	—	✓	text
SDTT (Deschenaux and Gulcehre 2025)	—	✓	✓	text
T3D (Zhang et al. 2026)	—	✓	✓	text
IMDM (Yoo et al. 2026)	—	✓	—	text
FMLM (Lee et al. 2026)	✓	—	—	text
SiFLOW (ours)	✓	✓	✓	text

yields an identity with no integral, exactly as in the continuous case.

Proposition 1 (Average–instantaneous consistency) For all $0 \leq s < t \leq 1$,

$$\mathbf{U}(\boldsymbol{\mu}_t, s, t) = \mathbf{v}(\boldsymbol{\mu}_t, t) - (t - s) \frac{d}{dt} \mathbf{U}(\boldsymbol{\mu}_t, s, t). \quad (3)$$

Equation (3) is the discrete-masked-diffusion analog of the MeanFlow identity; the only change from Geng et al. (2025) is the domain of $\boldsymbol{\mu}$. It exposes a one-step generator: integrating $\mathbf{v}$ over the whole interval is the same as a single average step.

Corollary 1 (One-step generation) With $s = 0, t = 1$, the displacement equals the average velocity, so

$$\hat{\boldsymbol{\mu}}_1 = \boldsymbol{\mu}_0 + \mathbf{U}_\theta(\boldsymbol{\mu}_0, 0, 1), \quad (4)$$

and tokens are read out per position by sampling or $\arg \max$ from $\hat{\boldsymbol{\mu}}_1$.

A computable target. We do not compute the integral. Two routes give a regression target for $\mathbf{U}_\theta$. (a) *Secant target*: run the teacher’s reverse sampler to obtain its predictive simplex points $\boldsymbol{\mu}_s$ and $\boldsymbol{\mu}_t$ at the two times, and regress onto the displacement quotient

$$\mathbf{U}_{\text{tgt}}(s, t) = \frac{\boldsymbol{\mu}_t - \boldsymbol{\mu}_s}{t - s}, \quad (5)$$

which equals the average velocity over $[s, t]$ exactly (it is the time-average of $\mathbf{v}$ by construction) and approximates the fixed-time instantaneous velocity $\mathbf{v}(\boldsymbol{\mu}_s, s)$ up to an $O(t - s)$ error (Proposition 2). (b) *Identity target*: substitute the parameterized $\mathbf{U}_\theta$ into the right-hand side of Eq. (3), replacing $\frac{d}{dt} \mathbf{U}$ by a Jacobian-vector product and stopping gradients through the target, exactly as in MeanFlow. We adopt the secant target (5) as the primary objective for its stability and low overhead, and treat the identity target as an optional refinement.

Remark 1 (Straight-path / flow-matching interpretation) In the affine coordinates of the simplex, the secant target $\mathbf{U}_{\text{tgt}} = (\boldsymbol{\mu}_t - \boldsymbol{\mu}_s)/(t - s)$ is the constant velocity of the straight segment $\boldsymbol{\mu}_r = \boldsymbol{\mu}_s + \frac{r-s}{t-s}(\boldsymbol{\mu}_t - \boldsymbol{\mu}_s)$, the geodesic of the flat (Euclidean) metric on the simplex, and exactly the straight-line conditional probability path of Flow

Matching (Lipman et al. 2023) restricted to the simplex. Learning $\mathbf{U}_\theta$ thus regresses the constant-velocity field of this straight path—the discrete-simplex analog of the straightened conditional paths that OT-Conditional Flow Matching (Tong et al. 2024) constructs to reduce inference steps. Two consequences follow. (i) The linear error scaling in Proposition 2 is the straightness dividend: a constant-velocity (zero-curvature) path is exactly integrable in one step, so the residual measures the curvature of the teacher’s simplex trajectory (Assumption 1, L). (ii) Cross-teacher distillation needs no architectural coupling: the target is a path in the shared output simplex, not in either model’s internal representation. We emphasize this is a flat-metric statement. A genuine W_2 optimal-transport reading would require a fixed ground cost $c(a, b)$ between vocabulary tokens; under such a metric the W_2 displacement interpolant generally differs from the Euclidean secant, and we leave a token-aware W_2 formulation to future work (Appendix B).

3.3 Simplex-Annealed Temperature Distillation

Single-pass generation must drive $\hat{\boldsymbol{\mu}}_1$ toward a near-vertex one-hot at each position: a near-zero-entropy target. Supervising this directly from the start is high-variance, because the masked-diffusion teacher is never asked to collapse a fully masked prior in one move—the documented *entropy collapse* problem (Zhu et al. 2025b). We address it with a temperature curriculum on the simplex, in the spirit of knowledge distillation (Hinton, Vinyals, and Dean 2015).

Let $\mathbf{z}_t$ be the teacher logits underlying $\boldsymbol{\mu}_t = \text{softmax}(\mathbf{z}_t)$, and let $\hat{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_s + (t - s)\mathbf{U}_\theta$ be the student’s implied endpoint. The *simplex-annealed temperature distillation* (SATD) loss is

$$\mathcal{L}_{\text{SATD}} = \sum_{i=1}^L \text{KL}(\text{sg}[\text{softmax}(\mathbf{z}_t^i / \beta)] \parallel \hat{\boldsymbol{\mu}}_t^i), \quad (6)$$

with a temperature $\beta = \beta(\text{step})$ annealed from $\beta_{\max} > 1$ (soft, high-entropy, easy to match) toward $\beta = 1$ (the teacher’s own sharpness) over training. Early high temperature gives the student a smooth target; late annealing restores the near-deterministic targets needed at $s \rightarrow 1$. The stop-gradient $\text{sg}[\cdot]$ keeps the teacher fixed. SATD is temperature-scaled knowledge distillation (Hinton, Vinyals, and Dean 2015) adapted to per-position simplex outputs; our contribu-

tion is the annealing curriculum that averts one-step entropy collapse, not the temperature mechanism itself.

3.4 Entropy-Injected Prior

The starting state for one-step generation is the fully masked prior, which carries no positional information and little entropy; a deterministic student then struggles to produce diverse, well-separated outputs (Zhu et al. 2025b). Following Di[M]O, we enrich the prior: with probability λ we replace a masked position’s input by a token drawn uniformly from $\mathcal{V}$, mixing a small amount of random signal into the prior,

$$\mathbf{x}_0^i \sim (1 - \lambda) \delta_{[\mathbf{M}]} + \lambda \text{Uniform}(\mathcal{V}), \quad \lambda \in [0, 0.3]. \quad (7)$$

This injects entropy at the input, relieving collapse pressure at the output while staying close to states the teacher saw during training. This prior is adopted directly from Di[M]O (Zhu et al. 2025b); we include it as a necessary ingredient, not a novel one. The SATD loss and the entropy-injected prior together make single-pass generation trainable: SATD provides the right difficulty curriculum, while the prior injection provides the distributional diversity.

3.5 Self-Conditioned Refinement

Reading out $\hat{\boldsymbol{\mu}}_1$ position-wise samples each token from its own marginal, so the one-step sample can contain individually likely but jointly inconsistent tokens—the token-independence failure. When a few extra passes are affordable, SiFLOW recovers dependencies cheaply: commit the highest-confidence positions of $\hat{\boldsymbol{\mu}}_1$, re-mask the rest, and re-apply the student at a short schedule of intermediate times $\{s_1 < \dots < s_{k-1}\}$. This is the standard masked-diffusion re-masking loop, but driven by the average-velocity student, so $k \ll$ the teacher’s step count. The one-step generator is the $k=1$ limit; we expect $k \in \{2, 4\}$ to be the practical quality-latency sweet spot, an expectation the protocol tests directly.

3.6 Training Algorithm

Algorithm 1 assembles the pieces. The student shares the teacher backbone and adds a two-layer velocity head predicting $\mathbf{U}_\theta$ (a displacement in logit space); only the velocity head is updated in the primary protocol. A light self-consistency term retains the teacher’s per-token accuracy on masked positions.

At inference, generation is Eq. (4) for one step, optionally followed by $k-1$ self-conditioned refinements (Section 3.5). No teacher is needed at test time, and the only inference overhead over a single MDM pass is the velocity head.

4 Theoretical Analysis

We relate one-step quality to how well the secant target tracks the teacher’s instantaneous velocity. Proofs are in Appendix A; here we state assumptions and results.

Assumption 1 (Regularity) *On the support of the teacher-induced flow, the instantaneous velocity $\mathbf{v}(\cdot, \tau)$ is L -Lipschitz in τ uniformly in $\boldsymbol{\mu}$, and the flow $\tau \mapsto \boldsymbol{\mu}_\tau$ is continuously differentiable.*

Algorithm 1 SiFLOW distillation (one training step)

Require: pretrained masked DLM teacher p_ϕ ; student $\mathbf{U}_\theta$ (teacher backbone + velocity head); prior-mix λ ; temperature schedule β ; weights λ_{reg} , boundary rate p_0

- 1: sample clean sequence $\mathbf{x}_1$ from corpus
- 2: with prob. p_0 set $s = 0$; else $s \sim \mathcal{U}[0, 1]$; sample $t \sim \mathcal{U}(s, 1]$
- 3: form **nested** masked states $\mathbf{x}_s, \mathbf{x}_t$ ($\text{mask}(t) \subseteq \text{mask}(s)$) at levels $\sigma(s), \sigma(t)$ from the same $\mathbf{x}_1$
- 4: teacher predictive simplex points $\boldsymbol{\mu}_s = p_\phi(\cdot | \mathbf{x}_s)$, $\boldsymbol{\mu}_t = p_\phi(\cdot | \mathbf{x}_t)$ {nested masks share a reveal order, so $\boldsymbol{\mu}_s, \boldsymbol{\mu}_t$ are a chord of one reverse trajectory; cacheable offline}
- 5: secant (OT) target $\mathbf{U}_{\text{tgt}} = (\boldsymbol{\mu}_t - \boldsymbol{\mu}_s) / (t - s)$; $\text{sg}[\mathbf{U}_{\text{tgt}}]$
- 6: student (entropy-injected prior) $\hat{\mathbf{U}} = \mathbf{U}_\theta(\boldsymbol{\mu}_s, s, t)$
- 7: $\hat{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_s + (t - s) \hat{\mathbf{U}}$
- 8: $\mathcal{L} = \mathcal{L}_{\text{SATD}}(\hat{\boldsymbol{\mu}}_t, \boldsymbol{\mu}_t; \beta) + \lambda_{\text{reg}} \mathcal{L}_{\text{MDM}}(\theta; \mathbf{x}_1)$
- 9: update θ via AdamW

Proposition 2 (First-order approximation error) *Under Assumption 1, the secant target (5) approximates the fixed-time instantaneous velocity at the interval start with*

$$\left\| \frac{\boldsymbol{\mu}_t - \boldsymbol{\mu}_s}{t - s} - \mathbf{v}(\boldsymbol{\mu}_s, s) \right\| \leq \frac{L}{2} (t - s). \quad (8)$$

The bound shrinks linearly in the interval, which is why a fixed network can serve all intervals and why the few-step variant (smaller intervals) is strictly easier to fit than the one-step limit. This mirrors the straight-path analysis: a perfectly straight path has $L = 0$ and zero gap between the secant and the instantaneous velocity, so the residual L measures the curvature of the teacher’s simplex trajectory. This yields a quality-versus-steps statement.

Proposition 3 (One-step gap) *Assume (A1) the student matches the average velocity in expectation over the masked prior, $\mathbb{E}\|\mathbf{U}_\theta - \mathbf{U}\|_1 \leq \varepsilon$; (A2) the residual entropy-collapse error of $\hat{\boldsymbol{\mu}}_1$ is at most η ; and (A3) the teacher’s clean predictive is bounded off the simplex boundary, $p_{\min} = \min_{i,a} \boldsymbol{\mu}_1^i(a) > 0$. Then the average per-position KL between the one-step generator and the teacher’s clean predictive satisfies*

$$\frac{1}{L} \sum_{i=1}^L \text{KL}(\boldsymbol{\mu}_1^i \| \hat{\boldsymbol{\mu}}_1^i) \leq \frac{(\varepsilon + \eta)^2}{p_{\min}}. \quad (9)$$

Corollary 2 (Monotone improvement with steps)

Partitioning $[0, 1]$ into k equal intervals and composing k average steps reduces the velocity-approximation term of Proposition 3 to $\varepsilon(k) \leq \frac{L}{2} \frac{1}{k}$ under Assumption 1, so the bound improves at rate $O(1/k^2)$ in the approximation-dominated regime.

These results explain the expected quality-step trade-off and motivate reporting one-step as the limiting case while few-step carries the headline number. The non-asymptotic form of Proposition 3 (via reverse Pinsker) and its proof are in Appendix A.

5 Experiments (Protocol)

Setup. This section describes the evaluation protocol. The setup, baselines, metrics, and ablations below are fixed, and the numeric cells in Tables 2–3 are auto-filled from the runs described in Table 4 (Appendix C). The protocol is deliberately scoped to a *single* A100-80GB GPU so that every reported number is reproducible without a cluster.

Models. The primary teacher is MDLM (Sahoo et al. 2024) ($\sim 170\text{M}$, DiT backbone, public OpenWebText checkpoint), chosen because it is small, open, and the canonical masked DLM. The student reuses this backbone and adds a two-layer velocity head ($\sim 1\text{--}3\text{M}$ trainable parameters; it operates in d_{model} space and lifts to logits via the frozen tied embedding, never a $K \times K$ matrix—see Appendix C), initialized from the teacher.

SiFLOW-D: Dream-7B teacher. Dream-7B (Ye et al. 2025) is a 7B-parameter open masked DLM initialized from Qwen2.5-7B ($\sim 14\text{ GB fp16}$), covering a qualitatively different training regime—AR initialization, instruction tuning, and a superior quality-speed tradeoff at 5–20 steps. The student shares the Dream-7B backbone and tokenizer, training only the velocity head; teacher targets are cached offline at ≤ 64 steps. SiFLOW-D validates cross-architecture generalization: the simplex-secant construction is agnostic to the teacher’s internals.

SiFLOW-G: DiffusionGemma teacher. DiffusionGemma-26B-A4B (Google DeepMind 2026) is a publicly released discrete diffusion LM (26B total, 3.8B active via sparse MoE; fp16 inference $\approx 7.6\text{ GB}$ active weights). The student is initialized from the DiffusionGemma backbone, on its native tokenizer, and adds only the velocity head. Teacher targets are cached offline. We designate this variant SiFLOW-G; it validates that SiFLOW generalizes to large, MoE-based teachers and produces a one-step generator at billion-parameter scale.

Data. We evaluate unconditional generation on OpenWebText (OWT) and LM1B, the standard masked-DLM benchmarks enabling like-for-like comparison with SDTT, FMLM, and IMDM. We additionally report LAMBADA zero-shot completion accuracy (Paperno et al. 2016), which probes the coherence of token dependencies—the primary failure mode of single-pass generation. Distillation targets are precomputed on a fixed corpus subset.

Baselines. (i) MDLM teacher at $\{1024, 64, 32, 8\}$ steps (quality ceiling and step-quality curve); (ii) BD3LM-small (Arriola et al. 2025) at 32 steps (discrete few-step reference with block structure); (iii) SDTT (Deschenaux and Gulcehre 2025) at 8 steps (best prior discrete distillation); (iv) T3D (Zhang et al. 2026) at 4–8 steps (concurrent trajectory self-distillation); (v) IMDM (Yoo et al. 2026) at 4–8 steps (concurrent few-step with infinite-state mask; trains from scratch, included to show one-step vs. few-step gap); (vi) a Di[M]O-style ablation, namely SiFLOW without the average-velocity target (pure one-step distribution matching), isolating our core contribution (Zhu et al. 2025b); (vii) FMLM (Lee et al. 2026), the continuous one-step language model, on matched OWT/LM1B settings; (viii) DLM-

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scripts/make_figures.py

Figure 3: **Simplex-trajectory straightness.** Per-sample path-length ratio $R = \int_0^1 \|\mathbf{v}(\boldsymbol{\mu}_\tau, \tau)\| d\tau / \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0\|$ for the teacher’s predictive simplex trajectories. $R \approx 1$ certifies that the path is nearly straight, validating the flat-geodesic premise of Remark 1 and the small- L regime of Proposition 2; large R would signal high curvature and a hard one-step limit.

pareto.pdf — generate via scripts/make_figures.py

Figure 4: **Quality-throughput Pareto frontier.** Gen-PPL (or MAUVE) versus throughput (tokens/s) for SiFLOW at $k \in \{1, 2, 4, 8\}$, the teacher’s step-quality curve ($\{8, 32, 64, 1024\}$ steps), and the baselines (SDTT, T3D, IMDM, FMLM, AR GPT-2). The velocity-head pass is included in all SiFLOW throughput numbers.

One (Chen, Zhang, and Zhou 2025) for the continuous one-step DLM comparison; (ix) an AR GPT-2 (124M) reference for latency and quality context (Radford et al. 2019).

Metrics. Generative perplexity (Gen-PPL) under a held-out GPT-2-Large scorer; MAUVE for distributional quality (Pillutla et al. 2021); unigram entropy and n -gram diversity to detect entropy collapse and mode collapse; LAMBADA zero-shot accuracy (chance = 0%) to test token-dependency coherence; NFE and wall-clock throughput (tokens/s) on the same single GPU; and, for the few-step variant, quality as a function of k .

Ablations and analyses. Beyond Table 3, we sweep the temperature schedule β , the prior-mix λ , and the refinement budget $k \in \{1, 2, 4, 8\}$; we plot the quality-throughput Pareto frontier against all baselines; and we visualize per-position entropy to confirm that SATD avoids the collapsed, overconfident distributions of the hard-label baseline. To evaluate token-dependency recovery, we report LAMBADA accuracy at each step count k , which should increase monotonically as the self-conditioned loop repairs jointly inconsistent samples.

Compute feasibility. Table 4 fixes the budget. The student is $\sim 170\text{M}$, trained in bf16 with AdamW at learning rate 1×10^{-4} on sequences of length 256 (primary) up to 1024 (secondary). Teacher targets are cached for a $\leq 2 \times 10^5$ -sequence subset with reverse rollouts capped at ≤ 64 steps. The full primary study fits in under 12 hours on a single A100-80GB.

6 Discussion and Limitations

SiFLOW rests on a single conceptual move—reading the flow map off the teacher’s predictive simplex—which both rebuts the “no discrete flow map” framing and keeps the method compatible with pretrained masked DLMs. The straight-path interpretation (Remark 1) explains why cross-

Table 2: Unconditional generation on OWT/LM1B. Arrows indicate better direction. “—” denotes a cell pending the runs in Table 4. Gen-PPL and MAUVE evaluated on OWT; LAMBADA is zero-shot accuracy. SiFLOW-D uses Dream-7B as teacher; SiFLOW-G uses DiffusionGemma-26B-A4B. †: number reported by the cited paper under its own protocol; all other rows measured by us on a single GPU. Tok/s and step counts include the velocity-head pass.

Method	Steps	Gen-PPL ↓	MAUVE ↑	LAMBADA ↑	Tok/s ↑
T3D [†] (Zhang et al. 2026)	4–8	—	—	—	—
IMDM [†] (Yoo et al. 2026)	4–8	—	—	—	—
FMLM [†] (Lee et al. 2026)	1	—	—	—	—
DLM-One [†] (Chen, Zhang, and Zhou 2025)	1	—	—	—	—
<i>Head-only distillation on larger backbones (each on its native tokenizer):</i>					
Dream-7B teacher (Ye et al. 2025)	~50	—	—	—	—
SiFLOW-D (ours)	1	—	—	—	—
SiFLOW-D (ours)	4	—	—	—	—
<i>DiffusionGemma backbone (head-only; A100-80GB):</i>					
DiffusionGemma (Google DeepMind 2026)	~1000	—	—	—	—
SiFLOW-G (ours)	1	—	—	—	—
SiFLOW-G (ours)	4	—	—	—	—
<i>Scale reference (different evaluation protocol):</i>					
LLaDA-8B (Nie et al. 2025)	~256	—	—	—	—

Table 3: Component ablation, 1-step generator. Each row drops one component from full SiFLOW.

Variant	PPL↓	MAUVE↑	LAMBADA↑	Tok/s↑
(run run_4 to populate)	—	—	—	—

architecture distillation is natural: because the target is a straight path in the shared output simplex, the student need not replicate the teacher’s internal representations—only track that path. This is what enables the same velocity head to distill MDLM, Dream-7B, and DiffusionGemma with identical training code.

Concurrent work IMDM (Yoo et al. 2026) independently arrives at a related geometric insight—it introduces an infinite-state mask that makes the factorization error bound arbitrarily small, and uses minibatch OT to couple training pairs. The key difference is one of where the geometry lives: IMDM applies OT coupling in *noise* space (and *replaces* the masking structure, requiring training from scratch), whereas SiFLOW regresses a straight path in the *output* simplex of any existing masked DLM. Whether the two approaches can be combined—using IMDM’s noise-space OT coupling to sharpen the simplex-space target—is a natural extension.

Several limitations bound the claims. The secant target is first-order (Proposition 2), so the one-step limit is expected to trail the few-step variant; we therefore report both and treat few-step as the headline. The self-conditioned refinement recovers token dependencies only approximately; a coupling-based sampler that enforces joint structure is a natural extension. The protocol is single-GPU by design, so it validates the mechanism at MDLM/170M scale and as head-only distillation on the larger Dream-7B and DiffusionGemma backbones (SiFLOW-D/-G); whether the one-step gap continues to narrow with student capacity is an

Table 4: Single-GPU compute budget. Primary targets <12 h on A100-80GB.

Phase	Config	Time
Cache targets	≤200k seqs, ≤64 steps	2–3 h
SiFLOW distil	30–60k steps, bs 32	6–8 h
Eval	5k samples, all baselines	1–2 h
Ablations	shared cache, head retrain	2–3 h
Total primary (single A100-80 GB)		<12 h
<i>SiFLOW-D (Dream-7B teacher)</i>		
Cache -D	Dream-7B fp16, bs 8	3–5 h
Distil -D	head-only, same protocol	6–8 h
Total SiFLOW-D (A100-80 GB)		<14 h
<i>SiFLOW-G (DiffusionGemma teacher)</i>		
Cache -G	3.8B active, fp16, bs 4	3–5 h
Distil -G	head-only, same protocol	6–8 h
Total SiFLOW-G (A100-80 GB)		<14 h

empirical question. A genuinely cross-tokenizer student—regressing one vocabulary’s simplex trajectory into another vocabulary’s simplex—is ill-posed without an explicit vocabulary alignment, and we leave it to future work; the variants here keep each student on its teacher’s native tokenizer. Finally, faster DLM inference lowers the cost of large-scale text generation, with the attendant dual-use considerations common to generative language models.

7 Conclusion

We presented SiFLOW, a framework for one- and few-step generation from pretrained masked diffusion language models. By defining an average unmasking velocity on the teacher’s predictive simplex, SiFLOW adapts the MeanFlow identity to discrete masked diffusion and supplies a flow-map analog where prior work assumed none. We showed

that the secant target is the constant-velocity field of a straight path on the predictive simplex—the discrete analog of the straightened conditional paths of Flow Matching and OT-CFM—explaining why cross-architecture distillation requires no architectural coupling, from any pretrained masked DLM, including Dream-7B and DiffusionGemma. A temperature-annealed distillation loss and an entropy-injected prior make single-pass generation trainable, and the construction reuses off-the-shelf DLMs rather than training from scratch. Our study covers three open DLM families on a single GPU; tightening the one-step gap toward the few-step result, and a token-aware W_2 formulation of the simplex target (Appendix B), are natural next steps.

A Proofs

Proof of Proposition 1. Fix s and define $G(s, t) = (t - s)\mathbf{U}(\boldsymbol{\mu}_t, s, t) = \int_s^t \mathbf{v}(\boldsymbol{\mu}_\tau, \tau) d\tau$, the second equality holding by the Definition of the average unmasking velocity. Differentiate both sides in t . On the left, the product rule gives $\frac{d}{dt}G = \mathbf{U}(\boldsymbol{\mu}_t, s, t) + (t - s)\frac{d}{dt}\mathbf{U}(\boldsymbol{\mu}_t, s, t)$. On the right, the fundamental theorem of calculus gives $\frac{d}{dt}\int_s^t \mathbf{v} d\tau = \mathbf{v}(\boldsymbol{\mu}_t, t)$. Equating the two and rearranging yields $\mathbf{U}(\boldsymbol{\mu}_t, s, t) = \mathbf{v}(\boldsymbol{\mu}_t, t) - (t - s)\frac{d}{dt}\mathbf{U}(\boldsymbol{\mu}_t, s, t)$, which is Eq. (3). The derivation is identical to the continuous MeanFlow identity (Geng et al. 2025); only the domain of $\boldsymbol{\mu}$ (the product of per-position simplices rather than Euclidean state space) differs. $\square$

Proof of Proposition 2. Because $\boldsymbol{\mu}_\tau$ is the flow of $\mathbf{v}$, we have $\boldsymbol{\mu}_t - \boldsymbol{\mu}_s = \int_s^t \mathbf{v}(\boldsymbol{\mu}_\tau, \tau) d\tau$, so the secant equals $\frac{\boldsymbol{\mu}_t - \boldsymbol{\mu}_s}{t - s} = \frac{1}{t - s} \int_s^t \mathbf{v}(\boldsymbol{\mu}_\tau, \tau) d\tau$. Subtracting the fixed-time velocity $\mathbf{v}(\boldsymbol{\mu}_s, s) = \frac{1}{t - s} \int_s^t \mathbf{v}(\boldsymbol{\mu}_s, s) d\tau$,

$$\left\| \frac{\boldsymbol{\mu}_t - \boldsymbol{\mu}_s}{t - s} - \mathbf{v}(\boldsymbol{\mu}_s, s) \right\| = \left\| \frac{1}{t - s} \int_s^t (\mathbf{v}(\boldsymbol{\mu}_\tau, \tau) - \mathbf{v}(\boldsymbol{\mu}_s, s)) d\tau \right\| \leq \frac{1}{t - s} \int_s^t L(\tau) d\tau = L \frac{t - s}{2} = \frac{L}{2}.$$

using the L -Lipschitz-in-time Assumption 1 for the integrand bound and Jensen’s inequality to move the norm inside the integral. Lipschitzness is well-justified here: the softmax read-out is C^∞ , the Transformer backbone is C^1 in its continuous inputs, and the composition $\tau \mapsto \boldsymbol{\mu}_\tau$ inherits this smoothness; the effective constant L is the curvature measured empirically by the path-length ratio of Figure 3. $\square$

Proof of Proposition 3. Fix a position i and write $p = \boldsymbol{\mu}_1^i$ (teacher clean predictive) and $q = \hat{\boldsymbol{\mu}}_1^i$ (one-step student). The student endpoint differs from the teacher target by the accumulated velocity error plus the entropy-collapse residual; in the ℓ_1 (total-variation) sense, $\|p - q\|_1 \leq \varepsilon_i + \eta_i$, where ε_i is the per-position velocity error and η_i the per-position collapse residual. By the reverse Pinsker inequality for distributions with $p \geq p_{\min}$ componentwise, $\text{KL}(p\|q) \leq \frac{1}{p_{\min}} \|p - q\|_1^2 \leq \frac{(\varepsilon_i + \eta_i)^2}{p_{\min}}$, using Assumption (A3). Averaging over positions and applying Jensen to $(\cdot)^2$ with $\frac{1}{L} \sum_i \varepsilon_i \leq \varepsilon$ (A1) and $\frac{1}{L} \sum_i \eta_i \leq \eta$ (A2) gives $\frac{1}{L} \sum_i \text{KL}(p\|q) \leq \frac{(\varepsilon + \eta)^2}{p_{\min}}$, the claimed non-asymptotic bound. Corollary 2 then follows: composing k equal-interval average steps drives the

velocity-approximation term to $\varepsilon(k) \leq \frac{L}{2k}$ by Proposition 2, so the bound contracts as the square of $\varepsilon(k) + \eta$. $\square$

B Straight-Path Proposition

Proposition 4 (Flat-metric geodesic on the simplex)

Equip the affine hull $H = \{\mathbf{x} \in \mathbb{R}^K : \mathbf{1}^\top \mathbf{x} = 1\}$ of the simplex with the flat (Euclidean) metric inherited from $\mathbb{R}^K$. For any $\boldsymbol{\mu}_s, \boldsymbol{\mu}_t \in H$ the unique constant-speed geodesic between them is the straight segment $\boldsymbol{\mu}_r = \boldsymbol{\mu}_s + \frac{r-s}{t-s}(\boldsymbol{\mu}_t - \boldsymbol{\mu}_s)$, $r \in [s, t]$, with constant velocity $\dot{\boldsymbol{\mu}}_r = (\boldsymbol{\mu}_t - \boldsymbol{\mu}_s)/(t - s)$. This is exactly the straight-line conditional probability path of Flow Matching (Lipman et al. 2023) restricted to H .

Proof of Proposition 4. H is an affine subspace, hence a totally geodesic flat submanifold of $\mathbb{R}^K$: the constraint $\mathbf{1}^\top \mathbf{x} = 1$ is preserved by any affine combination, since $\mathbf{1}^\top \boldsymbol{\mu}_r = (1 - \frac{r-s}{t-s})\mathbf{1}^\top \boldsymbol{\mu}_s + \frac{r-s}{t-s}\mathbf{1}^\top \boldsymbol{\mu}_t = 1$. Under the flat metric, geodesics are straight lines, and the straight segment minimizes Euclidean length among all curves joining the endpoints; differentiating it in r gives the constant velocity $(\boldsymbol{\mu}_t - \boldsymbol{\mu}_s)/(t - s)$, which is the secant target (5). Uniqueness follows from the strict convexity of squared Euclidean length. The Flow-Matching conditional path $\boldsymbol{\mu}_r = (1 - a_r)\boldsymbol{\mu}_s + a_r\boldsymbol{\mu}_t$ with linear schedule $a_r = \frac{r-s}{t-s}$ coincides with this segment. $\square$

Remark (token-aware W_2). The flat-metric reading above is geometry on the *coordinates* of the simplex and makes no reference to a cost between vocabulary tokens. If instead one equips the vocabulary with a ground cost $c(a, b)$ and reads $\boldsymbol{\mu}_s, \boldsymbol{\mu}_t$ as distributions over tokens, the W_2 displacement interpolant moves mass along c -optimal couplings. The two interpolants coincide only in the degenerate case of the discrete-uniform cost $c(a, b) = \mathbf{1}[a \neq b]$, for which W_2 reduces (up to a monotone transform) to total variation and the optimal coupling is the identity outside the mismatch set; for any nontrivial token metric (e.g. an embedding distance) the W_2 interpolant generally differs from the Euclidean secant. A token-aware W_2 formulation of the simplex target is therefore a genuine extension and is left to future work.

C Implementation Details

Velocity-head architecture. The velocity head operates in the teacher’s hidden dimension d_{model} , *not* in vocabulary space: it projects the backbone hidden state at each position through a two-layer FiLM-conditioned MLP (FiLM parameters are produced from the times (s, t)), and then lifts the d_{model} -dimensional output to a vocab-logit displacement through the *frozen* tied embedding $E^\top \in \mathbb{R}^{K \times d_{\text{model}}}$. The head therefore adds only $\sim 1\text{--}3\text{M}$ trainable parameters and—crucially—never instantiates a $K \times K$ matrix: the K -dimensional displacement is obtained by the frozen linear map $E^\top$, so the often-cited “a $K \rightarrow K$ MLP is billions of parameters” concern does not apply. The logit-space update keeps outputs on the simplex: $\hat{\mathbf{z}}_t = \mathbf{z}_s + (t - s)\mathbf{U}_\theta$ and $\hat{\boldsymbol{\mu}}_t = \text{softmax}(\hat{\mathbf{z}}_t)$, so the predicted endpoint is a valid categorical by construction.

Path consistency and prior. Secant pairs $(\boldsymbol{\mu}_s, \boldsymbol{\mu}_t)$ are formed from a single clean sequence under nested masks

entropy.pdf — generate via scripts/make_figures.py

Figure 5: **Per-position entropy** at $k = 1$ for the hard-label baseline, SiFLOW without SATD, and full SiFLOW. SATD prevents the collapsed, over-confident distributions of the hard-label baseline (Section 3.3).

lambda_vs_k.pdf — generate via scripts/make_figures.py

Figure 6: **LAMBADA accuracy versus refinement budget** $k \in \{1, 2, 4, 8\}$. Accuracy increases as the self-conditioned loop (Section 3.5) repairs jointly inconsistent tokens, consistent with Corollary 2.

$(\text{mask}(t) \subseteq \text{mask}(s))$, so the two states lie on one reverse trajectory (Section 3.1). The entropy-injected prior (Section 3.4) mixes a small fraction of uniform-token replacements into the masked input. For large-vocabulary teachers (Dream, DiffusionGemma) we regress on a reduced top- m support of the teacher predictive and fold the remaining mass into a single “rest” bucket, which keeps the head and the cached targets compact.

Hyperparameters. Unless noted: sequence length 256, batch size 64, AdamW with learning rate 2×10^{-4} and weight decay 0.01, 500-step linear warmup then cosine decay, 20k training steps. The SATD temperature anneals $\beta_{\max} = 2 \rightarrow 1$ on a cosine schedule over the first half of training. Loss weights are $\lambda_{\text{ent}} = 0.1$ (entropy prior), $w_{\text{vel}} = 0.1$ (velocity term), and $\lambda_{\text{reg}} = 0.05$ (MDM self-consistency). The prior-mix is $p_0 = p_1 = 0.25$, EMA decay is 0.999, and all reported numbers are averaged over seeds $\{0, 1, 2\}$.

D Extended Ablations

Beyond Table 3, we report: (i) a β -schedule sweep (anneal rate and $\beta_{\max}$), isolating the SATD curriculum from a no-anneal hard-label baseline; (ii) a λ prior-mix sweep over $\{0, 0.05, 0.1, 0.2, 0.3\}$ measuring the diversity/quality trade-off; (iii) head depth (1- vs. 2-layer MLP, and number of FiLM-conditioned mid-blocks); (iv) secant vs. identity target (Section 3.2); and (v) probability-space vs. logit-space heads. These correspond to the extended-ablation run group.

training_curves.pdf — generate via scripts/make_figures.py

Figure 7: **Training curves.** SATD loss and the MDM self-consistency term over training steps, used to diagnose collapse (flat curves) or instability.

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